

Exam Review

2025-09-18

Applied

Q1

Rapid antigen tests are commonly used to detect **COVID-19 infection**. Suppose a new rapid test is compared against the gold standard PCR test.

For this question, focus on **asymptomatic adults**. About **5% of asymptomatic adults** truly have COVID-19 infection. The test has a **sensitivity of 90%** and a **specificity of 95%** for detecting infection in this population.

- a. Suppose you administer the rapid test to someone who truly has COVID-19. What is the probability that the test will return a positive result? Circle one.

(A) 5%

(B) 90%

(C) 10%

(D) 95%

(E) Not enough information to determine this probability

- b. Suppose a randomly selected asymptomatic adult tests positive. Using Bayes' Rule, how could you **write out the probability that this person truly has COVID-19**?

- c. A new rapid test has been developed to detect **hepatitis B infection**. A study compares the results of the rapid test against the gold standard laboratory test. The results from 200 patients are summarized below:

	Gold Standard Positive	Gold Standard Negative	Total
Rapid Test Positive	70	25	95
Rapid Test Negative	10	95	105
Total	80	120	200

Question:

Calculate the **sensitivity** of the rapid test. Leave your answer as a fraction.

Q2

A researcher is studying the number of patients arriving at a hospital emergency room with asthma attacks during a 1-hour period. She proposes using a Poisson distribution to model the number of arrivals per hour.

Describe how you would verify whether each of the assumptions for a Poisson distribution are satisfied in this context. (e.g. what would you need to know/verify for each assumption, specific to this context?)

Q3

- a. An environmental agency classifies daily air quality in a city into three categories: Good, Moderate, and Unhealthy. The probabilities of each category on a randomly selected day are summarized in the table below:

Air Quality	Probability
Good	0.5
Moderate	0.3
Unhealthy	0.2

State the Kolmogorov axioms. Are all axioms true for this example?

- b. A public health researcher is studying the number of **asthma attacks requiring emergency care per day** in a mid-sized city. She models the number of attacks per day using a Poisson distribution with rate parameter $\lambda = 3$.

Recall the Poisson PMF:

$$P(X = k) = \frac{e^{-\lambda} \lambda^k}{k!}, \quad k = 0, 1, 2, \dots$$

Write out, in terms of the Poisson PMF, the probability that on a randomly selected day there are **at least 2 asthma attacks**. You do not need to compute the numeric value.

- c. Continuing part (b) above, Which of the following R commands correctly calculates the probability that on a randomly selected day there are **at least 2 asthma attacks**?

- (A) `dbinom(1, lambda = 3)`
- (B) `1 - ppois(1, lambda = 3)`
- (C) `ppois(2, lambda = 3)`
- (D) `1 - dpois(2, lambda = 3)`

Code

Elena Epidata is an epidemiologist exploring health trends in the U.S. population. She is using a dataset similar to **NHANES**, which includes variables such as **age, BMI, blood pressure, and cholesterol** for a sample of adults. Elena is new to R and is practicing basic data exploration.

Q4

Elena types the following code in R:

```
library(tidyverse)
nhanes <- read_csv("nhanes_sample.csv")
View(nhanes)
glimpse(nhanes)
names(nhanes)
```

Which of the following statements are true about the outputs of these commands? Select all that apply.

- (A) `View()` opens a spreadsheet-style view of the dataset.

- (B) `glimpse()` prints a concise summary of the dataset to the console.
- (C) `names()` lists the names of the columns.
- (D) `View()` returns a vector of column names.

Q5

Elena wants to examine a subset of the data. She types:

```
nhanes_highbp <- nhanes |>
  filter(age >= 50, sbp > 140)
```

- a. What subset of data has she created?
- b. What does `|>` do?
- c. Will anything be printed out by running this code?

Q6

Elena wants to create a scatter plot of BMI vs. systolic blood pressure and color points by sex. She writes:

```
ggplot(nhanes, aes(x = bmi, y = sbp, color = sex)) +
  geom_point() +
  labs(x = "BMI", y = "Systolic Blood Pressure (mmHg)")
```

Which of the following statements are true? Select all that apply.

- (A) `geom_point()` creates a scatter plot.
- (B) Points are colored based on BMI.
- (C) This code produces a boxplot of blood pressure.
- (D) The x-axis represents BMI.

Q7

Elena wants to calculate the average BMI by `age_group`. This variable has 3 categories: `young`, `mid`, and `older`. She types:

```
nhanes_summary <- nhanes |>
  group_by(age_group) |>
  summarize(mean_bmi = mean(bmi, na.rm = TRUE))
```

What does `nhanes_summary` contain?

Conceptual (True/False)

Circle either True or False.

Q8

The Poisson distribution is often used to model counts of rare events occurring independently in a fixed time or space interval.

True

False

Q9

In a discrete probability distribution, all individual probabilities must be greater than 0.

True

False

Q 10

Bayes' Rule allows you to compute the probability of an event given some observed evidence:
 $P(A|B) = [P(B|A) * P(A)] / P(B)$.

True

False

Q11

A frequentist interprets probability as a personal degree of belief, while a Bayesian interprets probability as the long-run relative frequency of events.

True

False